BPM2_ESOFT QUIZ 1_S2026_GD

# of Questions: 30

Question #: 1

A 23-year-old man is brought to the emergency department by the paramedics because of 1-day history of fever and abdominal pain. Diagnosis of acute appendicitis is made. Physical examination is most likely to show ?

A. Initial epigastric pain with radiation to the right upper quadrant
B. Initial umbilical pain with radiation to the left upper quadrant
C. Right lower quadrant pain with radiation to the umbilicus
✓D. Initial umbilical pain with radiation to the right lower quadrant
E. Initial suprapubic pain with radiation to the right lower quadrant

Rationale: Embryologically, the appendix is innervated by T10, as is the periumbilical region; thus, there is often referral of pain to this area initially in appendicitis. Visceral Afferent fibers will carry pain from the appendix to DRG
• With contact of the inflamed appendix with the body wall, pain is usually felt in the right lower quadrant. This pain will be Somatic Afferent fibers that will carry pain from the parietal peritoneum (body wall) to DRG

Question #: 2

A 65-year-old man is brought to the emergency room by the paramedics with a 1 hour history of vomiting bright red blood. He has had intermittent epigastric pain for the last 2 months. His medical history is significant for osteoarthritis for which he uses non-steroidal anti-inflammatory drugs daily. Physical examination shows epigastric tenderness. Imaging studies show a deep peptic ulcer located on the proximal aspect of the lesser curvature of the stomach. The ulcer most likely eroded which of the following arteries?

A. Splenic
B. Gastroduodenal
✓C. Left gastric
D. Common hepatic
E. Superior mesenteric

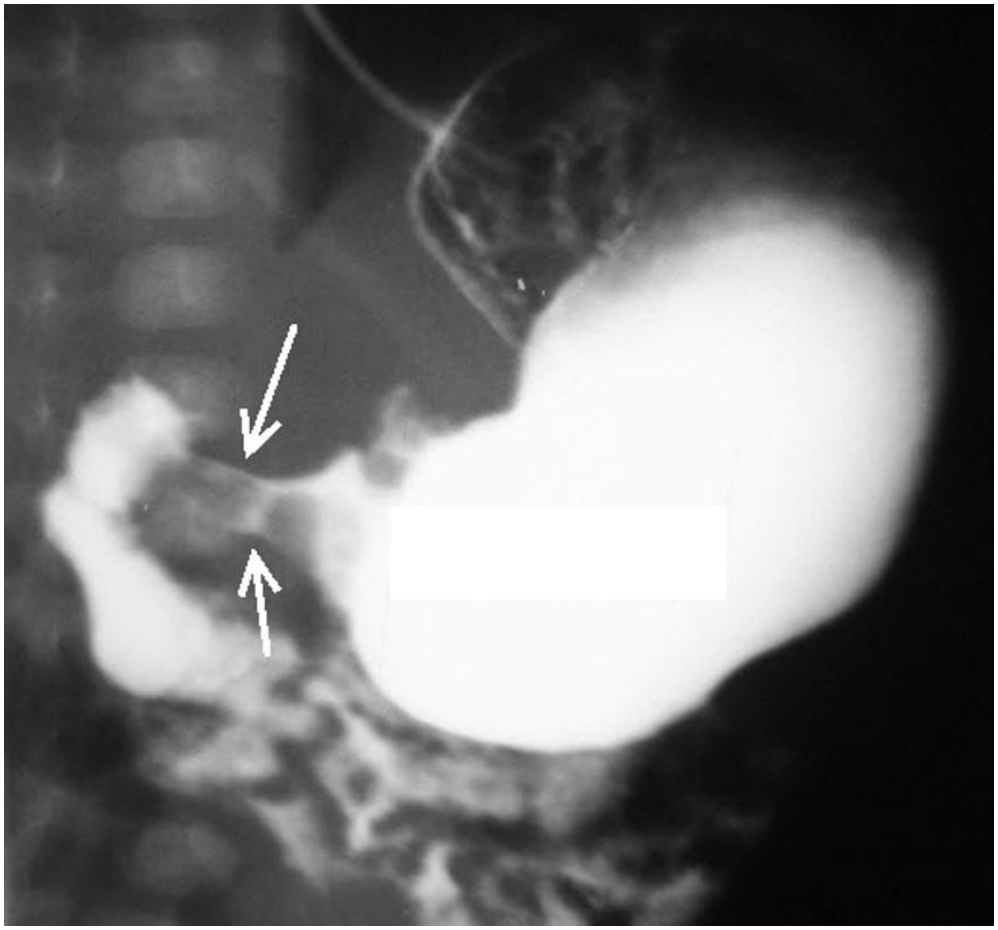
Rationale: Massive bleeding with hypovolemic shock is a potentially fatal complication of PUD. The majority of PUD cases are caused by H. Pylori infection or NSAID use. Other risk factors include smoking, glucocorticoid use, and older age. This patient has a risk factor of NSAID use. Most gastric ulcers arise on the lesser curvature of the stomach, usually at the transition zone between the body and antrum.
The left and right gastric arteries run along and perfuse the lesser curvature of the stomach and are a common source of hemorrhage from penetrating ulcers.

Question #: 3

A 4-month-old female infant is brought to the emergency department by her mother because of a 1-week history of projectile vomiting and poor feeding. Physical examination shows peristaltic waves moving from left to right and olive shaped mass was palpated in the epigastric region. A barium swallow is shown in the attached image. Which of the following most likely explains the clinical presentation?

A. Incomplete recanalization of duodenum
✓B. Narrowed pyloric canal
C. Complete occlusion of duodenum
D. Bifid ventral pancreatic duct
E. Narrowed proximal duodenum lumen

Rationale: Pyloric stenosis is the result of both hyperplasia and hypertrophy of the pyloric circular muscle fibers. The hypertro-phied pylorus can be palpated as an olive-sized mass in the right upper quadrant.



Question #: 4

A 58-year-old man comes to the physician because of a 2-week history of severe left sided abdominal pain. Imaging studies show a tumor on the posterior abdominal wall that has invaded into the superior mesenteric periarterial plexus. Which of the following structures is most likely to be affected?

A. Esophagus
B. Sigmoid colon
C. Stomach
D. Descending colon
✓E. Ascending colon

Rationale: Superior mesenteric periarterial plexus supplies the midgut structures.

Question #: 5

A 20-year-old man comes to the emergency department because of a 6-hour history of severe abdominal pain. Physical examination shows tenderness in the epigastric region. Imaging studies of the abdomen show an inflamed gallbladder. Which of the following is most likely the embryological origin of the affected structure?

A. Dorsal mesentery
✓B. Foregut
C. Midgut
D. Hindgut
E. Ventral mesentery

Rationale: The GI derivatives of the foregut are esophagus, stomach, duodenum up to the major duodenal papilla, gall bladder, liver and pancreas.

Question #: 6

A 45-year-old woman is admitted to the hospital for an elective cholecystectomy. During the procedure, a sharp object is accidentally stuck into the posterior boundary of the epiploic foramen. There is immediate bleeding. Which of the following structures is most likely the source of the bleeding?

A. Proper hepatic artery
✓B. Inferior vena cava
C. Portal vein
D. Right renal vein
E. Superior mesenteric vein

Rationale: The inferior vena cava is located directly posterior to the epiploic foramen and forms part of the posterior border. The hepatoduodenal ligament which contains the portal vein, common bile duct and the proper hepatic artery, forms the anterior border of the foramen.

Question #: 7

A 60-year-old man is brought to the emergency department by his wife because of a 4-hour history of severe abdominal pain. Physical examination shows abdominal tenderness and guarding. Endoscopic examination shows a perforated peptic ulcer located in the posterior wall of the duodenal cap. Which of the following arteries is most likely to be damaged due to this injury?

A. Splenic
B. Common hepatic
C. Proper hepatic
✓D. Gastroduodenal
E. Left gastric

Rationale: The gastroduodenal artery is located in the posterior aspect of the duodenal bulb, before it divides into the right gastro - omental and the superior pancreaticoduodenal arteries.

Question #: 8

A 1-day-old boy is under observation because of vomiting a large amount of bile-stained mucus. He has not passed meconium (first stools of a newborn). Physical examination shows his abdomen is distended. . An abdominal CT scan shows large dilated loops of bowel. A diagnosis of Hirschsprung disease is made. Failure of which of the following processes most likely resulted in this condition?

✓A. Migration of neural crest cells
B. Fusion of the lateral folds
C. Full rotation of the midgut loop
D. Withdrawal of herniated midgut loop into abdominal cavity

Rationale: Hirschsprung disease is due to lack of the myenteric and meissners plexus in the GI wall in the colon. These are derived from neural crest cells.

Question #: 9

A 56-year-old woman comes to the physician because of a 6-month history of weight loss and lethargy. She has experienced alternating bouts of constipation and diarrhea for 2 months and bloody stools for 1 month. Colonoscopy shows a tumor of the ascending colon. Which of the following lymph nodes will the tumor cells most likely initially metastasize to?

A. Celiac
B. Inferior mesenteric
C. Internal illiac

✓D. Superior mesenteric
E. Inguinal

Rationale: The ascending colon is part of the midgut. It receives blood supply from the superior mesenteric artery and is drained by the superior mesenteric veins. Therefore the lymphatic drainage is by the superior mesenteric lymph nodes.

Question #: 10

A newborn girl is examined by the physician a few hours after birth. Physical examination shows meconium (earliest feces) in her vaginal vestibule. Ultrasound of the pelvis shows anorectal agenesis with a rectovaginal fistula. Which of the following structures most likely developed abnormally in this patient?

A. Mesopnephric duct
B. Anal membrane
C. Cloacal membrane
D. Paramesonephric duct
✓E. Urorectal septum

Rationale: The urorectal septum divides the developing cloaca into the rectum and cranial part of the anal canal posteriorly and the urogenital sinus anteriorly. If this septum develops abnormally, there may be incomplete partition, resulting in a fistula connecting the rectum and vagina.

Question #: 11

A 22-year-old man is brought to the emergency department by the paramedics because of a 30-minute history of a gunshot wound to the anterior abdomen. A FAST exam (focused assessment with sonography for trauma) shows a large amount of fluid in the abdominal cavity due to damage to a major abdominal vessel. With the patient in the supine position, which of the following areas is fluid most likely to be found?

A. Rectovesical pouch
B. Paracolic gutters
C. Lesser sac
✓D. Hepatorenal pouch
E. Subphrenic recess

Rationale: When a patient is lying in the supine position, the lowest part of the abdominal cavity is the heptorenal recess. If this male patient was sitting or standing up, then the lowest part of the abdominal cavity would be the rectovesical pouch.

Question #: 12

A 62-year-old man comes to the physician because of a 6-month history of waking frequently in the night to urinate. Digital rectal examination shows an enlarged prostate. He is scheduled for a transurethral resection of the enlarged part of the gland. This type of procedure spares the parasympathetic nerves innervating the erectile bodies of the penis. Which of the following is most likely the location of the preganglionic cell bodies of these nerve fibers?

A. Wall of the organ
B. Superior hypogastric plexus
C. Inferior hypogastric (pelvic) plexus
D. Lateral horn L1-L2
✓E. Lateral horn like structures in S2-S4

Rationale: The cell bodies of the preganglionic parasympathetic nerve fibers that innervate the pelvic organs are located lateral horn like structures in S2-S4 segments of the spinal cord.

Question #: 13

A 75-year-old man is admitted to the hospital for surgery to repair a previously diagnosed abdominal aortic aneurysm. During the repair, the inferior mesenteric artery is usually excised. Which of the following arteries will most likely continue to supply blood to the distal colon through its anastomotic connections?

A. Ileal
✓B. Marginal
C. Common hepatic
D. Jejunal
E. Splenic

Rationale: Marginal artery is anastomosis between superior mesenteric and inferior mesenteric arteries.

Question #: 14

A 40-year-old man is brought to the emergency department by his wife because of a 2-day history of abdominal pain. Physical examination shows tenderness on palpation of the left lower quadrant. A CT scan of the abdomen shows multiple out-pouchings of the wall of the rectum leading to a diagnosis of diverticulitis. Which of the following nerve structures transmits the fibers that are responsible for the pain felt by this patient?

A. Lumbar splanchnics
✓B. Pelvic splanchnics
C. Greater thoracic splanchnics
D. Lesser thoracic splanchnics
E. Least thoracic splanchnics

Rationale: Pelvic splanchnic are parasympathetic fibers and carry pain fibers below pelvic pain line.

Question #: 15

An investigator is studying liver metabolism during a particular metabolic state when insulin: glucagon ratio is low and acetyl CoA levels are elevated. It is observed that acetyl CoA binds to an enzyme and allosterically activates it. Which of the following labeled hepatic enzymes is most likely being studied by the investigator?

A. Enzyme 1
B. Enzyme 2
C. Enzyme 3
✓D. Enzyme 4
E. Enzyme 5

Rationale: High levels of acetyl CoA acts as an absolute activator for Pyruvate carboxylase (Enzyme 4).
When the insulin: glucagon ratio is low as in the fasting state; Beta oxidation of fatty acids forms acetyl CoA;
Beta oxidation of fatty acids forms NADH and FADH2 and this reduces flux through the Krebs

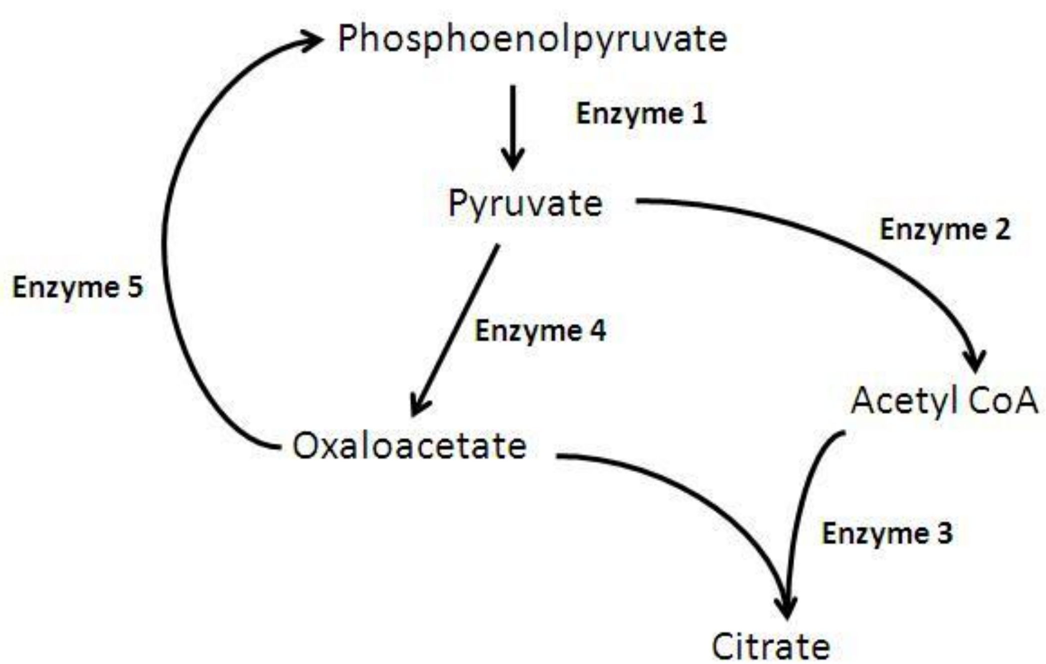
cycle during fasting.
Oxaloacetate from the Krebs cycle is being used for gluconeogenesis. Pyruvate is converted to oxaloacetate (by pyruvate carboxylase) which is used for gluconeogenesis. Note that this enzyme also requires Biotin as a coenzyme.

PDH complex (Enzyme 2) is inhibited by acetyl CoA, NADH. It is activated by calcium.

Oxaloacetate is converted to phosphoenolpyruvate by Phosphoenolpyruvate carboxykinase (PEPCK) - Enzyme 5 in the diagram - which is an irreversible reaction of gluconeogenesis. It is induced by cortisol.

Enzyme 1 is Pyruvate kinase (regulated enzyme of glycolysis) - Glucagon causes phosphorylation of the enzyme via cAMP -Protein Kinase A system and results in low activity of the enzyme. Pyruvate kinase is also inhibited by high Alanine levels in the fasting state.

Enzyme 3 is citrate synthase from the Krebs cycle; TCA cycle has low activity during fasting as Oxaloacetate is used for gluconeogenesis.



Question #: 16

A 25-year-old man comes to the physician because of a 2-month history of scaly skin and itching. Dietary history shows he eats raw eggs a part of his training for body building. Avidin in raw eggs inhibits absorption of the vitamin. Which of the following enzyme activities is most likely impaired as a result of this vitamin deficiency in this patient?

A. Phosphofructokinase-1
✓B. Pyruvate carboxylase
C. Pyruvate dehydrogenase complex
D. Glucose 6-phosphate dehydrogenase
E. Carnitine palmitoyl transferase-I (CPT-I)

Rationale: Avidin in raw egg white inhibits absorption of biotin.
Biotin deficiency results in low activity of carboxylases: Pyruvate carboxylase (gluconeogenesis), Acetyl CoA carboxylase (fatty acid synthesis) and propionyl CoA carboxylase (Conversion of propionyl CoA to methylmalonyl CoA --> Succinyl CoA).
PDH complex requires thiamine (vitamin B1), niacin (NAD+), riboflavin (FAD), lipoic acid and pantothenic acid (Coenzyme A).
Glucose 6-phosphate dehydrogenase requires NADP+ (niacin).

Question #: 17

A post-mortem examination is performed on a 48-year-old man who died due to suspected poisoning. Toxicology studies of hair samples show very high levels of arsenite (AsO_2^-). Which of the following reactions is most likely inhibited by this toxin?

✓A. Pyruvate to acetyl CoA
B. Citrate to isocitrate
C. Succinate to fumarate
D. Phosphoenolpyruvate to pyruvate
E. Fructose 6-phosphate to fructose 1,6-bisphosphate
F. Glucose 6-phosphate to glucose

Rationale: Arsenic poisoning (includes arsenate and arsenite) inhibits glycolysis (glyceraldehyde 3-phosphate DH), pyruvate dehydrogenase complex and alpha-keto dehydrogenase complex. Review the enzymes that catalyze the reactions listed.

Question #: 18

A 3-day-old boy is brought to the physician by his mother because of a 24-hour history of severe diarrhea. The mother says his stools are watery and foul smelling. Physical examination shows dehydration. An enzyme deficiency is identified. Which of the following bonds is most likely cleaved by this enzyme?

A. Alpha 1-6 glycosidic linkage
B. Alpha 1-4 glycosidic linkage
✓C. Beta 1-4 glycosidic linkage
D. Phosphodiester linkage
E. Beta N-glycosidic linkage

Rationale: The newborn has congenital lactase deficiency resulting in congenital lactose intolerance. Lactase hydrolyzes the beta 1-4 glycosidic linkage between glucose and galactose in the disaccharide lactose.

Question #: 19

A 45-year-old man comes to the physician because of a 6-month history of weight loss despite a good appetite. His current BMI is 22 kg/m2 and he has lost 10 pounds since his previous visit 6 months earlier. Laboratory studies show fasting plasma glucose is 180 mg/dL (Reference range: 70-100 mg/dL). Imaging studies show a tumor of specific cells of the islets of Langerhans. Laboratory studies show elevated levels of a specific hormone. Which of the following signal transduction proteins is most likely activated by this hormone?

✓A. Protein kinase A
B. Protein kinase C
C. cGMP-dependent protein kinase
D. Phosphodiesterase
E. Tyrosine kinase

Rationale: The vignette describes a patient with hyperglycemia due to increased secretion of a hormone from the islets of Langerhans; Glucagonoma is the most likely diagnosis. Insulinoma is characterized by hypoglycemic episodes and not hyperglycemia. The intracellular mechanism of action of glucagon is via activation of cAMP dependent protein kinase A

Question #: 20

A 55-year-old woman is brought to the emergency room by her friend in a delirious state. She has a 20-year history of alcohol abuse. Physical examination shows nystagmus and ataxia and a diagnosis of Wernicke-Korsakoff syndrome is made.

Which of the following enzymatic activities would most likely be impaired in this patient?

A. Pyruvate carboxylase
B. Acetyl CoA carboxylase
C. Glucose 6-phosphate dehydrogenase
✓D. α-Ketoglutarate dehydrogenase complex
E. Glutamate gamma-carboxylase
F. Succinate dehydrogenase
G. Cholecalciferol 1-hydroxylase

Rationale: Wernicke-Korsakoff syndrome is due to thiamine (vitamin B1) deficiency in chronic alcoholics. TPP is required as a coenzyme for transketolase, PDH and alpha-keto dehydrogenase complexes. It is also required for the branched keto acid dehydrogenase complex (branched chain amino acid catabolism)

Question #: 21

A liver biopsy is performed in a 50-year-old woman with a 3-month history of jaundice. The biopsy is performed after she had a carbohydrate rich meal and normal glycogen granules in the cytosol is observed. Which of the following statements best describes this macromolecule in the hepatocytes?

A. Degradation is allosterically activated by AMP via phosphorylase
B. Breakdown is stimulated by elevated blood insulin levels
✓C. Has an abnormal structure in patients with Andersen disease
D. Has β1-4 glycosidic linkages on the main chain and β1-6 bonds at the branch points
E. Synthesis is reduced at high glucose-6-phosphate levels

Rationale: Glycogen structure is abnormal in patients with Andersen disease. There is accumulation of glycogen with long branches due to deficiency of the branching enzyme. Glycogen has alpha 1-4 (main chain) and alpha1-6 (branch point) glycosidic links. Presence of glucose 6-phosphate increases glycogenesis. Glycogenolysis is stimulated when there is high levels of glucagon (low insulin: glucagon ratio). Muscle glycogen phosphorylase is allosterically activated by AMP (but the liver isoenzyme is not)

Question #: 22

A researcher observes that during strenuous, intense muscle contraction, a high energy compound directly forms ATP in the cytosol of white muscle fibers. Which of the following intermediates most likely performs this function?

A. Glyceraldehyde 3-phosphate
B. Succinyl CoA
✓C. Phosphoenolpyruvate
D. Glucose 6-phosphate
E. Fructose 1,6-bisphosphate
F. Pyruvate

Rationale: During strenuous muscle activity, there is low oxygen tension in the muscles. As a result, glycolysis occurs under anaerobic conditions. The substrate level phosphorylation reactions form ATP under these conditions. Pyruvate kinase and phosphoglycerate kinase form ATP. The substrates of these enzymes are the high energy compounds: Phosphoenolpyruvate and 1,3-bisphosphoglycerate.

Question #: 23

An investigator isolates the enzyme catalyzing the reaction shown. Which of the following statements best describes this enzyme catalyzed reaction?
Pyruvate →Acetyl CoA

A. Catalyzed by pyruvate carboxylase
B. Cytosolic enzyme
C. Requires pyridoxal phosphate as a coenzyme
D. Activated by low intracellular Ca^{2+} levels
✓E. Activated by dephosphorylation

Rationale: The enzyme catalyzing this reaction is PDH complex which is a mitochondrial enzyme. It requires TPP, CoA, FAD, NAD and lipoic acid as coenzymes. It is activated by high intracellular calcium levels and is active in the dephosphorylated state.

Question #: 24

A researcher studying skeletal muscle metabolism observes that glycogen degradation and glycolysis are closely regulated during muscle contraction. Increased concentration of intracellular calcium during muscle contraction results in activation of the calmodulin subunits and subsequent activation of an enzyme of intermediary metabolism. Which of the following enzymes is most likely activated by this mechanism during muscle contraction?

A. Glycogen phosphorylase
✓B. Glycogen phosphorylase kinase
C. Glycogen debranching enzyme
D. Hexokinase
E. Phosphofructokinase-1

Rationale: Glycogen phosphorylase kinase has calmodulin subunits which bind to calcium resulting in activation of glycogen phosphorylase kinase which in turn phosphorylates and activates glycogen phosphorylase hence activating glycogenolysis.

Question #: 25

A 38-year-old man was brought to the emergency room because of a 3-hour history of sweating, fever and restlessness. He works in a chemical factory and was accidentally exposed to 2,4-Dinitrophenol three hours earlier. Which of the following is the most likely mechanism of action of this toxin?

A. Inhibits electron flow through cytochrome oxidase
B. Decreases oxygen consumption
C. Inhibits ATP-ADP translocase
D. Decreases the rate of oxygen uptake by the mitochondrion
✓E. Facilitates proton movement into the mitochondrial matrix
F. Inhibits complex-II of the electron transport chain

Rationale: 2,4-Dinitrophenol is an uncoupler of oxidative phosphorylation. It facilitates proton movement into the mitochondrial matrix and dissipates the proton gradient across the inner mitochondrial membrane.

Question #: 26

A 27-year-old man is brought to the emergency department because he was found trapped in his house due to a wildfire. Physical examination shows a cherry red color of the nail beds. Which of the following components of the electron transport chain is most likely inhibited in this patient?

A. NADH dehydrogenase
B. Cytochrome reductase

✓C. Cytochrome oxidase
D. Fo-F1 ATP synthase
E. Succinate dehydrogenase

Rationale: Complex IV (cytochrome oxidase) of the ETC is inhibited by Carbonmonoxide. CO also binds to hemoglobin and reduces oxygen delivery to the tissues.
Other inhibitors of complex IV are hydrogen sulfide and cyanide. Review the inhibitors of each of the components of the ETC.

Question #: 27

A 2-year-old boy is brought to the physician by his mother because of a 3-month history of failure to thrive. Laboratory studies show hyperglycemia. Targeted gene mutation analysis shows a mutation in the gene encoding glucokinase. Which of the following best describes this enzyme?

✓A. Has a high Km for glucose
B. Used for the formation of glucose from lactate
C. Inhibited by glucose 6-phosphate
D. Fructose 2,6 bisphosphate is an allosteric activator of this enzyme
E. Induced by glucagon
F. Has a lower Vmax than hexokinase in muscle

Rationale: The comparison of kinetics of glucokinase and hexokinase: Glucokinase has a high Km, low substrate (glucose) affinity, and high Vmax. Hexokinase has a low Km, high substrate affinity and lower Vmax. Glucokinase does not show product inhibition by glucose 6-phosphate and is repressed by glucagon or starvation.

Question #: 28

An investigator studying liver metabolism recognizes that lactate produced in skeletal muscle is transported to the liver by the Cori cycle. In the hepatocytes, it is converted to glucose and released into the bloodstream. Which of the following hepatic enzymes is most likely involved in this pathway?

✓A. Phosphoenolpyruvate carboxykinase
B. Phosphofructokinase-2
C. Glucokinase
D. Glucose 6-phosphate dehydrogenase

E. Pyruvate kinase
F. Pyruvate dehydrogenase complex
G. Phosphofructokinase-1

Rationale: The conversion of lactate to glucose involves gluconeogenesis. The only enzyme involved in gluconeogenesis is PEPCK. The other possible answers could be pyruvate carboxylase, Fructose 1,6 bisphosphatase and Glucose 6-phosphatase. The other enzymes listed are irreversible reactions of glycolysis. G6PD is an enzyme of pentose phosphate pathway. PFK-2 is involved in the formation of fructose 2,6-bisphosphate, which is a modulator of glycolysis/gluconeogenesis.

Question #: 29

A 60-year-old woman comes to the physician because of a 3-month history of constipation and hard stools. She is advised to increase her dietary fiber content and to include a certain dietary polysaccharide. Which of the following best describes this polysaccharide?

A. Has α 1-6 and α 1-4 glycosidic bonds
B. Branched polysaccharide made of glucose
✓C. Unbranched homopolysaccharide with β 1-4 bonds
D. Digested by lactase in the small intestine
E. Digested by salivary and pancreatic amylase to form maltose

Rationale: Cellulose is a dietary fiber and is also a polysaccharide. Cellulose is an unbranched polysaccharide made of glucose units linked by beta 1-4 glycosidic linkages. It is not digested by the human digestive enzymes and contributes to roughage.

Question #: 30

The metabolism of 30 women volunteers aged 25-30 years is studied. Their liver metabolism is monitored 30 minutes after a breakfast of cereal and milk. It is observed that there is increased activity of hepatic phosphofructokinase-1 due to the presence of an allosteric activator. Which of the following intermediates is most likely functioning as an allosteric activator for this enzyme?

A. Glucose 6-phosphate
B. Fructose 6-phosphate
C. Fructose 1,6-bisphosphate

✓D. Fructose 2,6-bisphosphate
E. Glucose 1-phosphate
F. Acetyl CoA

Rationale: Fructose 2,6-bisphosphate formed by the action of PFK-2 is a positive allosteric effector of PFK-1. High levels of Fructose 2,6-bisphosphate inhibits the activity of Fructose 1,6-bisphosphatase (gluconeogenesis).

Fructose 1,6-bisphosphate is a feed-forward activator of pyruvate kinase.